

# THE STARSHIP GALILEO

## DEEP SPACE TRAVEL AND COLONIZATION

### INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

# PHASE 253

## THE TORPEDO POD (THE COURIER TORPEDO)

rescue or supplies for those we cannot turn for  
side-mounted — 5g normal to the human



the lunar throw at 2.4 km/s — the Moon, first service station

V1 LANE FERRY · V2 ORBIT TARGET · NO ROOF TREBUCHET

### THE POD GOES — THE SHIP HOLDS COURSE

Rescue logistics — v1.0 in force

### GP-IM-253

Revision v1.0 — 24 August 2026

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### VETTED BATCH 36 — TEST · STRUCTURAL AND HUMAN FACTORS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

## THE GEMINI PROTOCOLS — CHAPTER 253

**PHASE 253: THE TORPEDO POD (THE COURIER TORPEDO — RESCUE OR SUPPLIES FOR THOSE WE CANNOT TURN FOR; THE ARK NEVER DIVERTS) — STATUS: CURRENT — TEST CLASS / STRUCTURAL + HUMAN FACTORS + TRAJECTORY VALIDATION.** No extra single-level roof trebuchet on the roof at rear — the torpedo pod, with man, oxygen and food, sent as a rescue mission or full of requested supplies to someone we cannot turn around or off course for. The third never-divert law: the ark never diverts — not for logistics (251's mules run the ferry), not for reconnaissance (252's salvo), and now not for rescue: the pod goes, the ship holds course. Side-mounted so the g-force is normal to a human at 5g — the safety line, graded: correct as a PEAK for short burns, Apollo/Soyuz-class territory. The sling is a throw, not a gun: at the crewed 5g cap a ring gives  $v = \sqrt{a \cdot r}$ . Consumables about 1 kg of oxygen and 2 kg of food per man-day — a two-week pod carries about 100 kg. The lane envelope is huge: matching a 10,000 mph relative drift takes 91 seconds at 5g. **THE LUNAR THROW**, his spec: the launcher must throw the pod off the Moon — 2.4 km/s required — and the Moon becomes the Highway's first service station: shallow well, no air, free rocks for the bolo. V1 the lane ferry; V2 the orbit target, his pickup, seated for version 2 — brake into the field.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-253, v1.0 — 24 August 2026.

### §0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the pod law as seated (contents line, the design as spoken, the third never-divert law, the launcher physics, the pod and the man, the bench items) — §2 the physics, graded — §3 the system in the pod — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

### §1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 253: **The Torpedo Pod (The Courier Torpedo — Rescue or**

**Supplies for Those We Cannot Turn For; the Ark Never Diverts).** Single-level roof trebuchet at the rear roof (229's sling family) throwing a torpedo pod — man, oxygen, food, or requested supplies; side-mounted so g rides transverse at the 5g safety limit (peak, short burns; cruise 1-2g); soft sling for the man, hard sling for the cargo; lane-rescue round trips at 10-18 km/s  $\Delta v$ ; V1 the lane ferry, V2 the orbit target (well-rated, surface excluded by physics); the lunar throw: 2.4 km/s release off the Moon (10 km rail + coffee-cup pod burn, or the 251-family rotavator).'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

### THE DESIGN, AS SPOKEN (VERBATIM)

Archer: "no extra single level roof trebuchet on the roof at rear, torpedo pod with man oxygen and food, being sent as a rescue mission or just full of requested supplies to someone we cannot turn around or off course for, side mounting the pod so g force is normal to a human at 5g, the safety limit"

Archer: "for gravitational field i was thinking maybe even an orbit target, not everyone has a ship always up from a planet not even earth, nice pickup for V2 though"

Archer: "ok closer to home based on that 1969 tin can weighing over a ton and a coffee cup full of fuel, this must be able to throw this pod off the moon"

### THE DOCTRINE — THE THIRD NEVER-DIVERT LAW (VERBATIM)

- The ark never diverts — not for logistics (251's mules run the ferry), not for reconnaissance (252's salvo flies the look), and now not for people: **the torpedo goes instead**. Five hundred thousand tons does not maneuver for one man or one crate — the message is fired, not flown to. That is not cold; it is the only version of rescue that exists at fleet mass.
- **The boundary, seated so no reader overclaims:** a stranded giant belongs to 251's bolo; the far fast look belongs to 252's salvo; the pod ferries the near and the breathing. Three doctrines, one dovetail.

### THE LAUNCHER — THE PHYSICS NOTE (VERBATIM)

- **The sling is a throw, not a gun:** at the crewed 5g cap, a ring gives  $v = \sqrt{a \cdot r}$  — 10 m/s on 2 meters, 31 m/s on a 20-meter arm. The trebuchet's real jobs: clear the hull, orient the pod, and pay the first meters so the pod's own drive pays less. Two launch profiles, one launcher: **soft sling for the man (5g cap), hard sling for the cargo** — supplies feel nothing, so the same trebuchet can rip a cargo pod out at real speed and the pod rocket only finishes.
- Consumables: ~1 kg of oxygen, ~2 kg of food and ~4 kg of water per man-day — a two-week pod

carries ~100 kg of life. Nothing there.

- **The lane envelope is huge:** matching a 10,000 mph relative drift takes 91 seconds at 5g; a full lane-rescue round trip (go, stop, dock, come home) runs 10–18 km/s of  $\Delta v$  — a mass ratio under 1.5 on mule exhaust. The pod catches anything in the lane family, docks, and comes home.
- **THE LUNAR THROW (his spec, seated): the launcher must throw the pod off the Moon — 2.4 km/s release.** His benchmark honored: the 1969 tin can's ascent stage needed ~2 km/s of  $\Delta v$  on ~2.4 tonnes of propellant — not quite a coffee cup, but the Moon's well holds 1/22 of Earth's grip per kilogram, so the cup is directionally true. Three ways: (1) cargo rail — under 3 km of track at 100g, pocket change; (2) the crewed sweet spot — 10 km of rail at 5g for 1 km/s, the pod's rocket covers the remaining 1.4 km/s at a mass ratio of 1.03: the coffee cup, literally; (3) the rotavator — ~118 km arms at 5g crew comfort, 56% of the 20 GPa yarn ceiling, built from 251's own tether family — and it catches as well as it throws, so the sling runs on bookkeeping.
- **The Moon becomes the Highway's first service station:** shallow well, no air, free rocks for 251's stockpile, and a thrower that flings pods at escape speed for the price of electricity. The first bolo stockpile and the first pod launcher want the same address.

#### THE POD AND THE MAN (VERBATIM)

- **The 5g call, graded:** correct as a PEAK for short burns — Apollo/Soyuz-class territory — and the side-mounting is exactly right: g through the chest (eyeballs-in) is tolerable where g through the spine blacks you out. Clarification seated: 5g is minutes-class; the pod's working profile is 1–2g sustained, which a crew tolerates for hours to days, with 5g reserved for the punch burns.
- **V1 — THE LANE FERRY:** soft sling, pod burn, 1–2g cruise with 5g punch, weeks of air and food; catches anything in the lane family.
- **V2 — THE ORBIT TARGET (his pickup, seated for version 2):** well-rated — brake into the field, match orbit (~7.8 km/s around an Earth-class world), grapple, climb back out: ~20–30 km/s of budget, roughly triple V1, still inside mule-exhaust ratios; aerobrake where there is sky (the 252 shield-nose's cousin). **Surface excluded by physics, stated so no one oversells:** an Earth-class surface needs ~9 km/s of staged chemical climb — a lander, not a torpedo. V2's honest spec: rendezvous with anyone who can reach orbit — a station, a crippled ship, an escape pod — not anyone who can only reach the roof. His case recorded: 'not everyone has a ship always up from a planet not even earth' — Earth itself spent 2011–2020 unable to launch its own people on demand; if a planet-spanning civilization can have a crewed-launch gap, anyone can. The ark never diverts, and the well never gets a vote.

## **BENCH ITEMS — PHASE 253 (VERBATIM)**

(1) The 5g transverse couch and restraint spec (minutes-class punch, eyeballs-in). (2) The consumables pack standard (per man-day, two-week module). (3) Lunar rail vs rotavator trade study for the first service station. (4) Guidance — 81's intercept matrix heritage: the pod aims at moving targets by birthright. (5) Cargo-pod hardening standard for the hard sling.

## **ORIGIN OF THOUGHT — PHASE 253 (VERBATIM)**

Born from the 1-g game — 21 mph per second — and from a 1969 tin can that proved how shallow the Moon's well is. The lineage: 229's roof sling is the mother (a single-level trebuchet, rear roof, dedicated), 41's mules lend the drive, 252's salvo lends the shield-nose cousin, 251 lends the tether family for the rotavator option and the never-divert doctrine itself, 81 lends the aim. Locked by number on the author's word: '253'.

Build the Torpedo Pod: the last-resort battery — a pod of torpedoes for the situation nothing else resolves. Carried and prayed over in the hope it stays sealed. — INSTRUCTION RECONSTRUCTION (NOT VERBATIM — the original order was not preserved; reconstructed from the seated body, 10 August 2026, per sitting 146).

## **§2 — WHY IT WORKS (THE PHYSICS, GRADED)**

THE THIRD NEVER-DIVERT LAW, SEATED: the ark never diverts — not for logistics (251's mules run the ferry), not for reconnaissance (252's salvo), and now not for rescue. The torpedo pod is the third leg: someone we cannot turn around or off course for still gets the man, the oxygen, the food — the pod goes, the ship holds course. The boundary is seated so no reader overclaims: a stranded giant belongs to 251's bolo; the pod carries the small and the urgent.

THE LAUNCHER — THE SLING IS A THROW, NOT A GUN: at the crewed 5g cap, a ring gives  $v = \sqrt{a \cdot r}$  — 10 m/s on a small ring, scaling with radius; the roof-trebuchet amendment is his own kill: 'no extra single level roof trebuchet on the roof at rear' — the pod rides the existing throw machinery, side-mounted, and the launcher question is the phase's physics note.

THE 5G CALL, GRADED: correct as a PEAK for short burns — Apollo/Soyuz-class territory — with the g normal to the body by the side-mount. The human-factors item the vet named is the long-burn posture, and the phase owes it.

THE CONSUMABLES, PRICED: about 1 kg of oxygen and 2 kg of food per man-day — a two-week pod carries about 100 kg. The lane envelope is huge: matching a 10,000 mph relative drift takes 91 seconds at 5g — the pod is a ferry, not a chase vehicle, and the numbers say the ferry covers the lane.

THE LUNAR THROW — HIS SPEC, SEATED: the launcher must throw the pod off the Moon — 2.4 km/s

required. The Moon becomes the Highway's first service station: shallow well, no air, free rocks for the bolo line, and a throw that needs no atmosphere pricing. The shakedown's day trip is its first customer.

V1 AND V2, SEATED IN ORDER: V1 — THE LANE FERRY: soft sling, pod burn, 1-2g cruise with the 5g punch in reserve, weeks of air and food. V2 — THE ORBIT TARGET (his pickup, seated for version 2): well-rated — brake into the field. The version ladder is the doctrine: fly the ferry first, then rate the deep catch.

### §3 — THE SYSTEM IN THE POD

- **The law:** the ark never diverts — the third never-divert; the pod goes, the ship holds course.
- **The pod:** man, oxygen, food — about 100 kg for two weeks; rescue or requested supplies.
- **The mount:** side-mounted, g normal to the human, 5g peak cap — Apollo/Soyuz-class territory.
- **The throw:** the sling is a throw, not a gun —  $v = \sqrt{a \cdot r}$  at the 5g cap; no roof trebuchet, his kill.
- **The envelope:** 91 seconds at 5g matches a 10,000 mph drift — the ferry covers the lane.
- **The Moon:** 2.4 km/s off the lunar well — the Highway's first service station.

### §4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- The never-divert laws now stand at three: 251's mules for logistics, 252's salvo for reconnaissance, 253's pod for rescue — the ark's course is the law, and the exceptions are machines, not turns.
- 251's bolo owns the boundary: a stranded giant belongs to the tow truck; the pod's jurisdiction is the small and the urgent — seated so no reader overclaims either machine.
- 244's shakedown is the first customer: the Moon day trip puts the pod's launcher where the lunar throw is the day's work, and the service-station law gets its opening shift.
- The roof-trebuchet kill is his own amendment, seated verbatim: 'no extra single level roof trebuchet on the roof at rear' — the design deleted its own launcher before the audit could ask.

### §5 — VALIDATION: the grade and its full record live in the master and the restart.

### §6 — BUILD SEQUENCE

- 1. Rate the launcher:  $v = \sqrt{a \cdot r}$  at the 5g cap — ring radius versus throw speed, the lunar 2.4 km/s spec benched.
- 2. Fit the mount: side-mount, g normal to the human — the short-burn posture proven, the long-burn owed.

- 3. Stock the pod: about 1 kg oxygen and 2 kg food per man-day — the two-week 100 kg manifest sealed.
- 4. Fly the envelope: the 91-second 10,000 mph match demonstrated on the lane simulator, then on the lane.
- 5. Open the service station: the lunar throw as a shakedown task — the Highway's first service station opens with the day trip.

## §7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 251 (the bolo trail):** the boundary — the stranded giant is the bolo's job; the pod carries the small and the urgent; the never-divert laws divide the work.
- **Phase 252 (the encounter doctrine):** the sibling law — the salvo is the second never-divert; the pod is the third.
- **Phase 244 (the shakedown doctrine):** the first customer — the Moon day trip opens the service station; the lunar throw is the day's work.
- **Phase 229 (the sling):** the launcher family — the throw, not the gun; the pod rides the family's throw machinery at crewed caps.

## §8 — DO-NOT-BUILD

- Do NOT divert the ark — the third never-divert law: the pod goes, the ship holds course; rescue is a machine, not a turn.
- Do NOT build the roof trebuchet — his kill, seated: no extra single-level roof trebuchet on the roof at rear.
- Do NOT exceed the 5g cap — peak, short burns, g normal to the body; the long-burn posture is owed, not assumed.
- Do NOT size it for giants — a stranded giant belongs to 251's bolo; the pod's jurisdiction is the small and the urgent.
- Do NOT skip the lunar rating — the launcher must throw 2.4 km/s off the Moon; the service-station law depends on it.

## §9 — OWED

OWED: the vet's three, whole — structural validation of the pod and mount at the 5g peak; human-factors validation of the burn postures (short proven, long owed); trajectory validation across the lane

envelope and the lunar 2.4 km/s throw. Plus the two-week consumables manifest benched, not summed.

## **§10 — STATUS / PROVENANCE / CHANGE CONTROL**

STATUS: CHAPTER GP-IM-253, v1.0 — CURRENT — TEST CLASS / STRUCTURAL + HUMAN FACTORS + TRAJECTORY VALIDATION, seated law, master v2.1 (numerical print order). Built 24 August 2026, fourteenth of the 20-chapter 'ok next 20' shift (the author's order, 24 August 2026, seated in sitting 622). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618 and 622): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 253